BIDIPTA RANA

AM25M021 – M.tech Biomedical Engineering
Linkedin :- https://www.linkedin.com/in/bidipta-rana

Education

M.Tech in Biomedical Engineering **2025 - 2027**
Indian Institute of Technology, Madras, Tamil Nadu

B.Tech in Biomedical Engineering (Honors in Bioinformatics) | CGPA 10/10 **2021 - 2025**
Shri Ramdeobaba College of Engineering and Management, Nagpur, Maharashtra

Higher Secondary Education (HSC) | 93.17% **2019 - 2021**
Pace Junior Science College, Nerul, Maharashtra

Secondary School Education (SSC) | 95.6% **2019**
J H Ambani School, Lodhivali, Maharashtra

Work Experience

IIT Jodhpur | Research Intern **Jan 2025 – July 2025**

- Design and Development of Anthropomorphic Robotic Finger: A comparative study of the two types of actuation mechanism: twisted string actuation (TSA) and cable-pulling actuation (CPA). (The research article has been accepted for the December 2025 INACOMM International Conference on Machines and Mechanisms.)

BioSpectronics Pvt. Ltd. (Startup Company) | Research Intern **Oct 2023 - Oct 2024**

- Working on the development of a portable chemistry analyzer designed to detect various body analytes through concentration analysis of body fluids, enhancing diagnostic capabilities.

NIT Rourkela | Research Intern **May 2024 - June 2024**

- Designed and implemented a wireless device utilizing an ESP32 microcontroller to accurately measure hand-transmitted vibrations in power tools, enabling real-time data visualization at 600Hz.

MGM Medical College and Hospital | Biomedical Intern **May 2023**

- Acquired foundational knowledge of key biomedical instruments including health monitors, ECG machines, ventilators, and defibrillators, along with practical experience in their daily maintenance.

Academic Projects

Machine Learning based Robotic Hand using EMG Sensor
Developed a machine learning based robotic hand that accurately mimics natural hand movements using EMG sensors interfaced with a controller and servo motors, fostering advancements in assistive technologies.

Smart Glasses for Visually Impaired
Developed an assistive wearable device designed to enhance mobility and spatial awareness for visually impaired individuals. The system integrates ultrasonic sensors with a N-555 timer circuit to detect obstacles and provide real-time audio alerts through a buzzer.

Health Monitoring System using ESP32 and IoT
Developed a health monitoring system by interfacing ESP32 with a pulse oximeter, temperature sensor, and seven-segment display. Implemented IoT for remote monitoring, enabling real-time data visualization of vital health parameters.

Patents and Publications

Paper Publication

1. S. Palekar, B. Rana, A. Khaire, A. Awasthi, M. Mahule and J. Kalambe, "Microfluidic Paper-Based Portable Biochemistry Analyzer for Blood Parameter Sensing," in IEEE Sensors Journal, vol. 25, no. 10, pp. 18155-18162, 15 May15, 2025, doi: 10.1109/JSEN.2025.3558549.

2. A.Khaire, B. Rana, A. Awasthi, S. Palekar, M. Bhaiyya and J. Kalambe, "Graphite Electrode Based Bipolar Electrode

Electrochemiluminescence Portable Platform for Point-of-Care-Testing Applications," *2025 IEEE Applied Sensing Conference (APSCON)*, Hyderabad, India, 2025, pp. 395-398, doi: 10.1109/APSCON63569.2025.11144259.

3. J. B. Zalke, N. P. Narkhede, Dinesh Rotake, B. Rana, and C. Sharma, "Enzymatic Chemiresistive Interdigitated Electrodes Fabricated by Screen-Printing Method and Functionalized with Novel Green Synthesized Nanoparticles for Detection of Glucose Concentrations," *Lecture notes in electrical engineering*, pp. 505–519, Jan. 2025, doi: https://doi.org/10.1007/978-981-96-3758-443.

Design Patent: Automated Ironing Device (Design No. 412341-001, Published June 2024)
Designed an automated ironing device minimizing human interaction by integrating sensors and automated structures for precise ironing, pressure control, and movement.

Technical Skills

CAD Designing (Fusion 360), 3D printing, Python, C, Micro-controllers (ESP32, Arduino)

Research Skills

Technical Writing, Problem-Solving, Critical-thinking, Literature Review

---

Trainings and Seminars

3rd International IEEE Applied Sensing Conference (APSCON) **Jan 2025**
Presented the research paper "Graphite Electrode-Based Bipolar Electrode Electrochemiluminescence Portable Platform for Point-of-Care Testing Applications."

Hands-on Workshop on Electrochemical Biosensor Technology **Dec 2024**
Attended a hands-on workshop on Electrochemical Biosensor Technology, organized by Svatahh Bioinnovations, Gujarat Technology University, acquired theoretical and practical knowledge from experts in the field to enhance biosensor development skills.

5th International Clinical Engineering & Health Technology Management Congress **Nov 2023**
Participated in the 5th International Clinical Engineering & Health Technology Management Congress, held at AMTZ, Visakhapatnam, networking with industry professionals and gaining insights into innovative healthcare technologies.

Achievements and Certifications

GATE 2025: AIR 82 in BM
IBM: Python for Data Science
L&T Edutech: Data Analytics Fundamentals

Positions of Responsibility

Vice Chairperson, IEEE EMBS RCOEM Student Branch: Coordinated technical events and student engagement activities, promoting collaboration in biomedical engineering initiatives .

Executive Member, REEF RCOEM: Organized and managed social initiatives including food drives, manjha collection drives, nature trail etc.